

MAINS MASTER PROGRAM (MMP) 2024

INTERNAL SECURITY – 5;

CYBER SECURITY

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2. PYQS

- What are the different elements of Cyber Security? Keeping in view the challenge in cyber security, examine the extent to which India has successfully developed a comprehensive National Cyber Security Strategy (Answer in 250 words)
- Keeping in view India's Internal Security, analyze the impact of cross-border cyber attacks. Also discuss defensive measures against these sophisticated attack [10 marks, 150 words]
- Discuss the types of Cybercrime and measures required to be taken to fight the menace [10 marks, 150 words] [Mains 2020]
- What is CyberDome project? Explain how it can be useful in controlling internet crimes in India. [10 marks, 150 words][Mains 2019]
- Discuss the potential threat of cyber-attacks and the security framework to prevent it. [10 marks, 150 words][Mains 2017]
- Considering the threats cyberspace poses for the country, India needs a "Digital Armed Force" to prevent crimes. Critically evaluate the National Cyber Security Policy, 2013 outlining the challenges perceived in its effective implementation. [200 words, 12.5 marks] [Mains 2015]
- Cyber warfare is considered by some analysts to be larger threat than even Al-Qaeda or terrorism. What do you understand by Cyber Warfare? Outline the cyber threats which India is vulnerable to and bring out the state of the country's preparedness to deal with the same. [200 words, 10 marks] [CSM 2013]
- What is a malware? [50 words] [CSM 2007]
- What is a firewall? [50 words, CSM 2007, CSM 2002]
- Who are 'hackers' in computer world? [50 words, CSM 2006]
- What is VIRUS? [50 words, CSM 2005]
- What is an 'Internet Worm'? Explain DDoS [CSM 2003]
- How does Computer virus destroy data? [CSM 2000]
- What is computer virus? How does it infect systems?[CSM 1999, 5 marks]

3. OTHER PRACTICE QUESTIONS

- Hardware Cyber Security Concerns is becoming another major threat for India's Cyber Security. Discuss [12.5 marks, 200 words]
- Lack of cyber trained workforce is a key concern of India's Cyber Security Strategy. Suggest some measures to remedy the problems. [10 marks, 150 words]
- "Cyber deterrence, on the line of nuclear deterrence, can dissuade cyber attackers" Critically Analyze [10 marks, 150 words]
- What is a ransomware? Suggest some measures to protect your system from attack by a ransomware. [10 marks, 150 words]
- Write a short note on:
 - » DNS Flood
 - » Malware
 - » Ransomware
 - » Advanced Persistent Threat
- Differentiate between Internet worm, Virus, Ransomware.

4. SOCIAL MEDIA AND INTERNAL SECURITY

- **Introduction:** Social media has seen remarkable growth in India in recent years
- **Key factors increasing the penetration of social media in the country:**
 - » Increasing penetration of internet; Affordable smart phones; Young Demography
 - » More content in local language
 - » Social media is also seen as a source of news, entertainment etc.
 - » COVID-19 pandemic accelerated the adoption of digital technologies, including social media. Social media was seen as a way to remain connected and informed during the covid restrictions.
- **Key threats from social media to internal security of the country:**
 - » **Fake news and hate speech** - social media can easily be used to spread fake news and hate speech which may harm communal harmony and increases polarization in society.
 - Deep fakes have further made it easy to propagate false information.
 - » **Tool of radicalization and recruitment**: Several terror organizations have used social media to spread their ideology, plan attacks and recruit new members.
 - Apps like 'Telegram' makes it difficult for security agencies to take action.
 - » **Cyber-attacks and data breaches**: Many first-time social media users are not aware of the risk associated with social media and thus fall prey to cyber-attacks, data breaches, honey trapping etc.
 - » **External interference in internal affairs**: social media has been used a tool of interference in democratic affairs of the country.
 - » **Mobilization for unlawful activities**: While social media has helped people to group together, it has also allowed anti-social elements to better organize and associate for illegal and anti-social activities.
- **Steps by India to prevent the misuse of social media:**
 - » Information Technology (Intermediary guidelines and Digital Media Ethics Code) Rules 2021
 - » Section 67 of Information Technology Act (IT), 2000 punishes transmission of obscene material in electronic format
 - » PIB's fact check unit under IT Rules 2021
- **Way Forward:**
 - » **Strengthening of laws and policies and their effective implementation**: Rigorous laws against fake news, cyber bullying, cybercrime and their effective implementation.
 - » **Technology**: Stay upto date with latest advancements and use the latest technologies like AI and ML to strengthen the ability of authorities to fight challenges emanating from social media.
 - » **Awareness Generation**: People having social media accounts need to be made aware of general challenges emanating from social media and how they can be prevented. They should also know how to report harmful content, block abusive accounts etc.
 - » **International cooperation**: Setting up international standards which will balance the privacy vs security debate and provide authorities with access to identity of people involved in illegal activities.
- **Conclusion:**
 - » By adopting an integrated and holistic approach, focusing on above dimensions, India can mitigate the risk associated with social media and ensure a safe and secure online environment for all users.

5. ISSUE OF ONLINE RADICALIZATION

- **Introduction:** With the increase in penetration of Internet and social media, there has also been a **sharp rise in extremist rhetoric**. This specially threatens countries which are multilingual, multicultural, and religiously plural.
- **Key factors contributing to online radicalization:**
 - » **Echo Chamber and Filter Bubbles:** The algorithms used by social media repeatedly expose users to the content which reinforces their belief and biases. This may lead to radicalization of users.
 - » **Allows targeting based on region/ religion etc:** Users from different parts of the world can be connected and targeted through social media by radical organizations. They may employ persuasive techniques like emotional appeal, promises of belonging etc.
 - » **Anonymity** of online spaces may allow individuals to promote extremist content without fear of repercussions.
 - » **Victimization Narratives** - These can be propagated easily by extremist organizations to attract followers. For e.g. targeting youth by showing problems faced by people in Gaza.
 - » **Oratory and Charisma:** Leaders of extremist organizations may also attract followers through their oratory and charisma.
- **Major challenges in curbing online radicalization:**
 - » Increasing penetration of internet, smartphones and social media.
 - » **Encrypted apps** like telegram makes it difficult to track illegal activities by authorities.
 - » **Gamification techniques** being used to spread ideology
 - » **Weak Laws** - lack of comprehensive definition of radicalization, terrorism etc.
- **Key initiatives by India to curb online radicalization:**
 - » **Unlawful Activities (Prevention) Act (UAPA), 1967** has been amended to include provisions for prosecuting individuals involved in online radicalization and recruitment for terrorist activities.
 - » **Information Technology Act, 2000** empowers government to block websites, social media accounts etc. to control radicalization
 - » **National cyber security policy** aims to strengthen cyber security and counter online threats, including threats from online radicalization.
 - » **Various initiatives for Digital Literacy and awareness**
 - » **Initiatives like "Sahi Raasta Program"** - it is a significant initiative by Indian army in collaboration with J&K administration to counter radicalization and bring vulnerable youth back into the mainstream.
- **Way forward:**
 - » Strengthening legal and regulatory framework:
 - » Enhancing human resources - the current police force and cyber security teams are lacking understanding of cybercrimes and key online threats.
 - » Enhancing technological capabilities - Law enforcement agencies need to be equipped with latest technologies to track and counter online radicalization
 - » Expanding awareness and outreach - India needs to launch a mass awareness campaign for cyber Swatchta ensuring common citizens are able to see the signs of radicalization.

- » **Enhancing international cooperation** - Since the source of online radicalization can be in any country, it is important to coordinate and cooperate with other countries to create a global framework against radicalization and take joint action against people involved

6. HONEY TRAPPING IN CYBER WORLD

- **Example Questions**
 - » "Honeytrap is a weapon of hybrid warfare being waged by the enemy across the borders" Discuss with examples. [12.5 marks, 200 words]
 - » The age-old spy-craft has been reinvented for digital age in the form of digital honey trapping. Suggest measures India should take to deal with these threats. [10 marks, 150 words]
- **Introduction**
 - » In the world of intelligence, information is the principal currency that the enemy is trying to acquire. In the various methods used by enemies, **spying using sex or lure of sex is one of the oldest tricks**. This mechanism is known as honey trapping.
 - » But, with higher penetration of internet, the art of honey trapping has also turned digital. Spies use mechanisms like infecting the lives and devices of the target from social media or injecting malware on someone's device (phone or computer) through adult sites.
 - According to reports, 3 of the world's top 20 viewed sites are porn related. Similarly, 25% of android malwares are related to porn.
- **Modus Operandi for honey trapping people?**
 - » Like the photographs by a soldier on social media. leave a comment like "thank you for keeping us safe", "Jai Hind" etc., initiate a conversation and eventually move to intimate messages over Whatsapp, Telegram etc.
- **Steps being taken to deal with honey trapping**
 - » Armed forces have issued **advisories to security forces** to beware of honey trapping.
 - Army has cracked a whip on social media usage with a list of do's and don'ts.
 - » The **Military Intelligence** is also carrying out selective checks on phones, laptops, and desktops of officers and soldiers in sensitive areas.
 - » An **information warfare team** has been set up at the Army Headquarters.
 - » **Suspected twitter handles** and **Facebook accounts** have also been identified.
- **Way Forward/ Other Steps that can be taken?**
 - » **Technological Changes**
 - **Invest in latest technologies** for early and better detection of viruses.
 - **Conducting timely reviews and audits** of all devices.
 - » **Procedural Steps**
 - **Developing improved Protocols/SOPs** in case of contamination
 - Developing a methodology to embed dormant malware in all sensitive data and devices which will be able to track the bad actors and destroy the documents with a programmed kill switch.
 - » **Awareness Generation**
 - **Sensitize defence personnel** against cyber risks
 - **Better cyber practices** must be built amongst fresh recruits.
- **Conclusion**

- » With enhancement in digital penetration of all the sectors, the modern warfare is becoming more and more faceless. Unlike past, the cost and barrier to entry into enemy territory has gone down drastically.
- » In this information age, the enemy will be relentless and continue to invest and recruit heavily in new methods. India needs to act fast to deter such threats.

7. DISINFORMATION AS A CYBER SECURITY THREAT

- **Disinformation is an attack and compromise of our cognitive being.** It uses manipulated, miscontextualized, misappropriated information, deep fakes and cheap fakes.
- **Cognitive hacking** is a threat from disinformation and computational propaganda. It exploits psychological vulnerabilities, perpetuate biases, and eventually compromises logical and critical thinking, giving rise to cognitive dissonance. The goal is to manipulate the way people perceive reality.
 - » The **storming of US Capitol** by right-wing groups on Jan 6, 2021, is a prime example of the effects of cognitive hacking.
- **Information ecosystem may be managed** by Nation-state actors, ideological believers, violent extremists, and economically motivated enterprises to achieve:
 - » **Violates Fundamental Right to Speech and Expression**
 - » **Create Social tensions and increase polarization**
 - » **Hampers healthcare efforts** - For e.g. COVID-19 disinformation campaign had prevented people from wearing masks, using potentially dangerous alternative cures, and not getting vaccinated, making it even more challenging to contain the virus.
 - » **Affect democratic process** - influence the outcome of an election.
 - » **Hampers Economic Growth:** The advertisement centric business modes and attention economy incentivize malicious actors to run a sophisticated disinformation campaign and fill the information channels with noise that drown the truth with unprecedented speed and scale.
 - » **Impact Diplomatic relations** - Relation with foreign countries
- **Deepfakes** add a whole new level of danger to disinformation campaigns.
- **Way Forward**
 - » **Develop Disinformation defence systems** by studying strategy and tactics to understand the identities of malicious actors, their activities, and behaviours from the cybersecurity domain to mitigate disinformation threats.
 - » **Develop a defense in depth strategy** (a multilayered strategy):
 - **Authenticity and provenance solutions** can intervene before the information gets posted.
 - **Detection solution** using human and artificial intelligence, internal and external fact checking can label or remove the content.
 - » **An integrated platform to identify disinformation creators** could be crucial to fight disinformation.

- **Currently**, the response to disinformation is in silos of each platform with little or no coordination.
- » **Invest in media literacy efforts** to reach out to discerning public. Media education can make big difference in understanding context, motivations, and challenging disinformation to reduce damage.
- » **Concerted efforts by all stakeholders** - Governments, businesses, non-governmental organizations, and other entities to create standards and implement defences.

8. CYBER SECURITY: BASICS

1. Basics

i. Introduction

- Cyberspace is the connected Internet Ecosystem". Today it covers almost all crucial defence and civilian infrastructure including IT network, banking and finance, transportation, water supply etc.
 - It has also emerged as the largest unregulated and uncontrolled domain in the history of mankind.
 - It is the **fifth potential theatre of warfare** along with land, sea, air and space.
 - Thus, protecting cyber space has become a very high priority for every government in the world.
- **Cyber Crime** is a criminal activity on the cyberspace.
 - It involves offences committed against people/ organization/ governments with a criminal motive to cause physical or mental harm, or loss to victim directly or indirectly.
 - It also threatens nation's security and financial health.
 - Cyber criminals are getting sophisticated with each passing day and are wreaking havoc by defacing websites and stealing confidential and privileged data for financial gain.
- **Cyber Security** is protecting our cyber space from attack, damage, misuse and economic espionage.
 - It is a complex issue that cuts across multiple domains and calls for multi-dimensional, multilayered initiatives and responses.

1) GENERAL CYBER SECURITY CHALLENGES:

Ensuring cyber security (unlike border security or military security) is very challenging because of the following factors:

1. **Cyberspace has inherent vulnerabilities** that can't be removed
 - **Innumerable entry points** to internet: Even if a few points are vulnerable, the whole system can be cyber attacked. The coming up of technologies like IoT which connects various 'things' with internet- increases the vulnerabilities of internet.
 - **Increasing penetration of Internet** with a large number of first-time users.

- **Increasing connectivity of different sectors** -> more data being stored digitally -> more vulnerability.
- 2. **Difficult to identify the source of attack:** For e.g. CIA's UMBRAGE project have advanced capabilities of misdirecting attribution to another nation-state ("**false flag attacks**") by leaving behind false fingerprints.
- 3. **Fast evolving technologies:** This allows cyber attackers to quickly exploit vulnerabilities. Further in the absence of timely updates, attack technologies surpass the defense technology allowing exploitation of the vulnerabilities.
- 4. **Nation states, non-state actors, and individuals** are at a **peer level**, all capable of waging attacks.
- 5. **State Sponsored:** Some countries **promote, shelter and encourage** cyber-crime as a political tactic.
- 6. **Cyber criminals (hacking groups have become very organized)**
 - They are part of the underground ecosystem that channels tools, expertise, and infrastructure in criminal operations that extract billions of dollars of profit from data theft, extortion, and fraud.

2) WHAT DAMAGES CAN CYBER ATTACK MAKE?

- **Attack on National Security:**
 - **Cyber Espionage, Warfare, Terrorism** etc. has emerged as a new challenge in India's National Security.
 - Cyber-attacks on India originating from China has increased manifold after the increased tensions on the borders.
 - For e.g. According to a US Cyber Security and intelligence firm, Recorded Future, Chinese state sponsored actors may have deployed malware into Indian power grids and sea ports as border tension rose in 2020 along the LAC.
 - US-Israeli joint efforts had leashed the Stuxnet Worm in 2010 - which helped it disable several hundred centrifuges at the Iranian nuclear facility in Natanz.
 - **Disabling of Critical Infrastructure** such as telephone network, satellite system, financial services can be brought by Cyber-attack.
 - For e.g. the **malware 'DTrack'** had breached a system at Kudankulam Power Plant.
- **Economic Loss - Financial Frauds**
 - Cybersecurity Ventures estimate the global annual cybercrime cost to be \$6 trillion in 2021.
 - **Information Loss:** The largest cost impact from cybercrime is information loss (39%).
 - **Business Disruption:** Loss (36%)
- **Information Warfare:** It aims to impact the decision-making process of the target's citizenry for advancing particular narratives.
 - With Big data, Behavioural and predictive analytics and AI, the nature of warfare in cyberspace has evolved.
 - **Sixth Generation Warfare (6GW)** applies 'reflexive control' aided by these developments to target in an individualized manner and change the beliefs of the leadership or citizenry of the adversary through mass data and behavioral analytics.

- **Social Media** has been used for:
 - **Fake news** - promoting social tensions and harming national integrity
 - **Radicalization** of youth has been happening through online platforms.
 - **Honey trapping**
 - **Data phishing**
- **Cyber Hacktivists:** They hack sites and servers to virally communicate the "message" for specific campaigns.
- **Violation of Right to Privacy:** Individual's personal data is disclosed on public platforms.
- **Other Crimes:** e.g. Online Abuse, Harassment, Stalking, Child Pornography etc.

3) CYBER SECURITY SITUATION IN INDIA

- **World Cyber Crime Index (2024)** ranks India **10th as major cybercrime hotspot**.
- **Cybercrime increased drastically** after the **increased border tensions with China**.
 - » Most of these attacks are DDOS, phishing, data exfiltration, remote access tool malware and keylogging (tracking every keystroke made by a user).
 - » Government websites and web portals have been the targets in the past too. What is different this time is that attackers appear to be well planned, and aimed at extracting information and sensitive data.

4) LAWS, POLICIES, SCHEMES, INSTITUTIONS ETC. TO ENSURE CYBER SECURITY IN THE COUNTRY.

- **Legal Framework**
 - Currently, the **Information Technology Act, 2000** is the primary law for dealing with cyber-crimes in India.
 - Section 43 (Data Protection), Section 66 (hacking), Section 66B punishment for illegally possessing stolen computer resource and communication devices, Section 67 (protection against unauthorized use of data), Section 69 (Cyber terrorism), etc. are some key provisions which cover Cyber Security.
- **National Cyber Security Policy 2013**
 - **Aim:**
 - The policy aims at
 - i. **Protection** of information infrastructure in cyberspace
 - ii. **Reduce vulnerabilities**
 - iii. **Build capabilities** to prevent and respond to cyber threats
 - iv. **Minimize damage** from cyber incidents through a combination of institutional structure, people, process, technology and cooperation.
 - **Key Provisions**
 - a. **Institutional Framework to promote Cyber Security**

- **National Critical Information Infrastructure Protection Centre (NCIIPC)**: it is a 24X7 mechanism to deal with cyber threats and works under NTRO to safeguard critical infrastructure.
- **CERT-IN** has been designated as nodal agency for coordination of crisis management. It will also coordinate and operationalize sector CERTs.

- Developing Cyber-security Human Resource** (creating a workforce of 5,00,00 professional in 5 years)
- Promote R&D**: More Investment; Industry-Academia collaboration; Setting up Centre of excellence in cyber security areas; R&D in cutting edge security tech; focus on indigenous development of cyber security solutions.
- Promoting PPP and collaborative engagements** through technical and operational co-operation
- Each organizations to promote cyber security through an information security policy and using only certified IT products**.
 - A person should be designated to work as Chief Information Security Officer (CISO);
 - Every organization should use certified IT products in sensitive security areas.
- Other provisions include** Updation of legal framework; Awareness program on cyber space; Periodic review of the adequacy and effectiveness of cyber security infrastructure etc.

- **Need of Review**

- **Technological advancements** in the field of Artificial Intelligence, 5G, Internet of Things etc. have made the 2013 policy outdated.
- **Human Resource Development** has been poor
- **Need of Governance Reform**
 - India has 36 different central bodies to look after cyber issues. Each organization has its own reporting structure and CERT (Indian Computer Emergency Response Team). This **plethora of agency** leads to confusion.
- **Need of a Unified Cyber Security Framework** across various regulators It is time to put a central command on CBI or CEC's lines, which will be a single point authority at mid-level.
- **Need of a Cyber-Defence Agency** - to implement the cyber defence strategy for national security.
 - This should also provide for cyber commando forces to neutralize any cross border cyber terrorism or cyber-attack and help in development of specialized cyber police cadres in all state police departments.
- **Awareness level** in police and judiciary is still very low about cyber security.

- **Joint Doctrine for Cyberspace Operations (2024)**: Released by India's Chief of Defence Staff, is a landmark document that provides a comprehensive framework for India's military to operate in the cyberspace domain.

» Recognizes cyberspace as the fifth domain of warfare

- **Indian Cyber Crime Coordination Center (I4C) - A 7-pronged Scheme to Fight Cyber Crime:** A threat analysis unit, a reporting portal, platform for joint cybercrime investigation team, improving laboratory ecosystem, focusing on training centre;
- **Institutions**
 - » **National Security Council (NSC)** usually chaired by the NSA plays a key role in shaping India's cyber policy ecosystem.
 - **National Information Board**, also chaired by NSA, is meant to be the apex body to promote cross-ministry coordination on cybersecurity policymaking.
 - » **National Critical Information Infrastructure Protection Centre** established under the NTRO in Jan 2014 is mandated to facilitate the protection of critical informational infrastructure.
 - » The **Indian Computer Emergency Response Team (CERT-IN)** is the national nodal agency which responds to various cyber security threats to non-critical infrastructure.
 - » Ministry of Defence (MoD) has established a **Defence Cyber Agency**, a tri-service command of the Indian armed forces to coordinate and control joint cyber operations and craft India's Cyber doctrine.
 - » The **National Cyber Coordination Center (NCCC)** is India's operational cybersecurity and e-surveillance agency, its main role is in screening communications metadata and coordinating intelligence collection among agencies.
 - » **Cyber Swatchta Kendra (Botnet Cleaning and Malware Analysis Centre)**: It has been launched for detection of malicious software programmes and to provide free tools to remove the same.
- **Cyber AASHVAST: India's first Cyber Crime Prevention Unit**
- **Other Initiatives to Promote Cyber Security In India**
 1. **Awareness Generation:**
 - **Cyber Surakshit Bharat Initiative:** Launched in 2018 to promote awareness about cyber crimes and building capacity for safety measures for Chief Information Security Officers (CISOs) and frontline IT staff across all government departments
 - **PMGDISHA**
 2. **Cyber Swacchta Kendra:** This a new desktop and mobile security solution for a secure cyber space in the country. It provides information and tools to users to secure their systems and devices.
 3. All the new government websites and applications are audited prior to hosting and on regular basis after hosting.
 4. **Increased International Cooperation**

5) PROBLEMS ASSOCIATED WITH INDIA'S CYBER SECURITY SYSTEM

1. **Cyber Security Policy, 2013 and IT Act** needs to be updated to ensure that it remains fast tracked.

2. **No Procedure Code for the investigation of Cyber or computer-related offences:** As electronic evidence is entirely different in nature when compared to evidence of traditional crime, laying down standard and uniform procedures to deal with electronic evidence is essential.
3. **Lack of coordination among the various institutions involved in cyber security:** We don't have a unified national cyber security architecture. There are too many agencies which have led to **concerns around effective coordination, overlapping responsibilities and lack of clear institutional boundaries and accountability**.
4. **Lack of trained manpower / Lack of focus on cyber security in educational and research institutes:**
 - » In June 2023, a report by TeamLease Digital said that India had 40,000 job openings in the field of cyber security, but 30% of these vacancies couldn't be filled due to huge skill shortage.
 - » The 2013 policy envisaged 5,00,000 experts in 5 years, but we had reached only 10% of the target.
 - So, most of the government offices completely lack the cyber security expertise required and thus are very vulnerable to cyber threats.
 - Further, there is a lack of technical staff for the investigation of cybercrime.
5. **Dependency on ICT Imports**
 - » **Main hardware attacks** can be **Manufacturing Backdoors; Hardware tempering** etc.
 - » Further, **lack of digital solutions** like data-sharing facilities and social media platforms have adversely impacted nation's self-reliance and cyber security.
6. **Lack of awareness among public** regarding the cyber security and significance of it.
 - » According to a survey by security solutions provider **NortonLifeLock**, around 52% adults admitted that they don't know how to protect themselves from cybercrime.
7. **Rapidly changing technology:** This makes most of our infrastructure outdated within a few years. This also requires continuous training of our manpower without which they also become redundant. All this leads to poor infrastructure and manpower fighting against cyber threats in the country.
8. **Use of Substandard Devices and rampant use of unlicensed software** increases cyber security vulnerabilities in India
9. **Lack of private investment**
10. **India's cyber capabilities are mostly defensive and lacks offensive power**
 - » A cyber arsenal is also needed for strategic deterrence, and this will also enhance our defensive approach.
11. **Almost Complete absence of Cyber Security Framework at state level** in many states
12. **Absence of 'Global Cyber Norms'** which can balance the competing demands of national sovereignty and transnational connectivity.

6) CYBER DETERRENCE

- Cyber security analysts have been debating whether cyber deterrence, on the lines of nuclear deterrence, can dissuade cyber attackers.
- Cyber deterrence can be of two types
 1. **Defensive** - By denying attack
 - Cyber defences are raised so that the attacker is unable to pierce the adversary's networks
 2. **Offensive** - By Punishment

- In this case, the cyber attackers are assured of a devastating response
- **Evidently, deterrence doesn't work in cyber space**
 - Attackers have been found to bypass the best of cyber defences
 - And for offensive cyber deterrence, it is necessary to identify the attackers with pinpointed accuracy. But attribution has proved to be Achilles heel of offensive cyber deterrence.
 - Indiscriminate targeting can prove to be more destabilizing and counter productive.
- **Some analysts believe that for cyber deterrence to be effective, the response need not always be in cyberspace**
 - The response can be economic, political or military.
- **Way forward for India**
 - The Indian military needs to make a proper assessment of an offensive cyber doctrine adopted by many countries and undertake action that goes beyond simply the building of defensive capabilities.

7) WAY FORWARD AND CONCLUSION

- **Reform policy, laws** - Both Cyber Security Policy 2013 and the IT Act, 2000 don't fully sync with today's cyber threat.
 - » The new cyber security policy should also ensure coherence and coordination between various institutions.
 - » It's high time that data privacy laws are prioritized by the legislature.
- **Human Resource Development: Skill shortage has to be dealt with high priority:**
 - » Fill all the relevant position with cyber security experts.
 - » States should set up a special cyber police station in each district or range, having technically qualified staff in every police station.
- **Upgrade Cyber Labs:**
 - » Cyber forensic labs of states must be upgraded with the advent of new technologies.
 - » The state of art, **National Cyber Forensic Lab** and the **Cyber prevention, Awareness and Detection Centre (CyPAD)** of the Delhi Police should also be asked to extend professional help to state labs.
- **Focus on Infrastructure and More R&D in the sector**
 - » Focus on how AI and ML can boost cyber defenses.
 - » Promote cutting edge technology growth in the country to reduce dependency on imports of ICT.
- **Enhancing Cyber Warfare Capabilities:**
 - » These enhancements would be technological, organizational, and human, employed for cyber offence, cyber defence, cyber deterrence, or combination of these.
 - » Developing **offensive deterrence capabilities**

- Create a **Cyber Defence Agency** which will specialize in neutralizing cyber attacks from across the border.
- **Specialized cyber security cells** are needed in police departments of each state.
- **Promote Cyber Hygiene:**
 - » **Regular updates of hardwares and softwares**, regular password changes etc.
 - » **Ensure security in imported ICT devices**
 - Set up **Trust Centres** (labs) to determine the level of trustworthiness of ICT vendors.
 - » **Nurturing young netizens through digital civics**.
- **Reduce dependency on imported ICT devices:** Promote ICT manufacturing in the country.
- **Promote the role of Private Players**
 - » **Encourage startups** in the field of social networking, cloud computing etc.
 - » Private sector should get involved in **more industry focused processes** such as the Microsoft-initiated Cybersecurity Tech Accord and the Siemen's led Charter of Trust.
- **Increasing International Cooperation to develop Cyber Norms.**
 - » According to **Budapest Convention, or Convention on Cybercrime of the Council of Europe** (CETs no. 185), which started as a European Initiative but has attracted others, is an option that India should examine.
 - » Further, International agreements should provide a binding obligation on 'red lines' with respect to cyberspace-targets that should be considered illegitimate due to their significance for human life, such as health-care systems, electricity grids, water supply and financial systems.
- **Prepare for implication of Quantum Computing on Cyber security.**
- **Protect the country from informational (6th Generation warfare):** - Measures to reduce foreign data collection; sectoral data localization; Enact the data protection bill; Integrate this modern form of warfare while developing India's security strategy.

Conclusion

- In India, it is imperative for our national security that cyber networks, software and cyber-physical systems, and platforms should be cyber-secure. This requires a judicious mix of people, policies and technologies, as well as robust Public Private Partnership.
- Various institutions need to be strengthened to bring in a **much-needed synergy** among various institutions and to work out a coordinated approach to cyber security, including cyber deterrence.

9. CYBER SECURITY THREATS FROM CHINA – CHINESE HARDWARE AND SOFTWARE

- **Example Questions:**
 - » "Adequate measures need to be adopted to strengthen India's cyber defence as it has become increasingly susceptible to Chinese Cyber Attacks" Elaborate [10 marks, 150 words]
- In 2018, CERT-IN (Computer Emergency Response Team - India) reported that China was responsible for 35% of the total number of cyber attacks on official Indian website. Military forces suspect that this attack has increased substantially after the Galwan clashes.
- **Recent Example of Chinese Cyber Attacks:**
 - » **Cyber-attack on AIIMS** in Dec 2022 which may have compromised data of 3-4 crore patients.
 - » **Attack on Ladakh Power Grid** in April 2022.
 - » Earlier in 2021, Chinese state-backed hackers' group had targeted the IT system of two Indian vaccine makers - Bharat Biotech and Serum Institute of India (SII)
 - » The Oct 2020, Mumbai power outage was also a result of multiple malware deployed by Chinese group RedEcho.
- **Various types of Cyber threats from China:**
 - » **Direct Cyber-attacks** - Hacking, DDOS, Phishing, etc.
 - » **Threats from Chinese apps and hardware**
 - » **Smart Technology also pose security risks:** SMART Products which encompass the range of everyday technology that is being operated across residential and office spaces in India. These include CCTVs, air conditioners, refrigerators, coffee machines, printers etc. Even the western produced devices depend on data sensors, modules and transmitters.
 - In UK, a report sent to government explains that these Chinese components can be used to track the movements of intelligence officers and ministers. It can also be used for stifling industrial activities.
- **Key steps taken:**
 - **Ban on several Chinese apps and technology** due to data leaks, vulnerabilities, and national security risks that they pose
 - » In 2020, India started banning Chinese applications which now cover more than 250 applications.
 - **Examples of Chinese firms under wider scrutiny in India**
 - » Exclusion from 5G telecom trials
 - » Increasing restrictions on research collaboration
 - » Income Tax Searches
- **Way Forward:**
 - **Work on outlining technical evidence** to prove that the attacks are from China.
 - **Complete analysis of Chinese Products** (Apps, Software, SMART Products) etc being used in India.
 - **Work on developing competitive replacements** to Chinese firms
 - **Expand Cyber Defence and Cyber Offensive Capabilities** to counter these cyber-attacks.

- **Other suggestions to strengthen Cyber Security** (already discussed in detail)

10. HACK FOR HIRE SERVICES – ITS IMPLICATIONS

- **Hack-for-Hire** refers to groups or individuals who sell unlawful computer access services, often to wealthy clients.
 - Unfortunately, India is developing a reputation as a hotspot of the hack-for-hire industry.
- **Lack of Strict Action against Hack-for-Hire firms in India:**
 - For e.g. US prosecutors indicted Sumit Gupta, the Director of Indian hacking firm BellTrox in 2015 for hacking on behalf of two American lawyers, yet the Indian government never took action against him.
 - This has encouraged the hack-for-hire industry in India to grow.
- **International Reactions:**
 - **Major partners** have not yet shown concerns for India's domestic hack-for-hire players, there are signs of tension.
 - **For e.g:**
 - **The Citizen Lab** at the University of Toronto recently highlighted the Indian hack-for-hire industry into the spotlight with a 2020 report that exposed the hacking group **Dark Basin** (linked to BellTrox, a hack-for-hire firm in India). This group targeted NGOs based in USA that spoke against oil and gas conglomerate ExxonMobil.
- **Does India have a legal framework to punish them?**
 - Yes, the Information Technology Act, 2000, and its subsequent amendments in 2008 specifically criminalized unauthorized computer usage and identity theft through electronic means along with other cybercrimes.
- **So why no punishment?**
 - Lack of political will.
- **Negative Impacts:**
 - Failing to address its own hotbed of domestic cybercrime threatens its relationships with other countries, especially given that these groups target foreign nationals. Further, it jeopardizes its role as an upcoming leader in global cybersecurity.
- **International Collaboration in fight against Cyber Security:**
 - India is party to **numerous international cybersecurity partnerships**.
 - In addition to multilateral partnerships like the International Multilateral Partnership Against Cyber Threats (IMPACT) and QUAD, India has also entered into a number of bilateral partnerships with countries like Israel, Egypt, Maldives etc.
- **Way Forward and Conclusion:**
 - India can build on the existing initiatives (like Operation Chakra) with countries like Australia and USA and take advantage of this cooperation to bring hack-for-hire firms to Justice. Such efforts

would encourage future relationships by showing global partners that India is **serious about cybersecurity**, both domestically and abroad.

- **If not reined in**, the hacking industry could impact India's cybersecurity strategy, and therefore its national security strategy as well.

11. STATE'S USE OF NON-STATE ACTOR IN CYBER SPACE

- **News:**
 - » Without proper legislation, states using non-state actors for cyberwarfare pose a significant threat to governments, businesses, and individuals (Aug 2023)
- **Introduction:**
 - » Creating a national cyber force is a time consuming and costly affair. Therefore, many states are turning towards non-state actors to bridge their shortcomings.
 - » **A Cyber non-state actor** is an entity with no specific physical territory or territorial sovereignty that operates in cyberspace. It can include individuals, groups, or organizations that operate independently or in association. Non-state actors can significantly threaten governments, businesses, and individuals.
- **Why do states choose non-state actors?**
 - » **Protect themselves:** Though cyberspace ensures anonymity, non-state actors provide additional protection to the states during cyberattacks, as they can **claim plausible deniability** and avoid blame to evade indictment.
 - For instance, North Korea uses **Bureau 121**, a hacking group, to carry out cyberattacks primarily against South Korea while maintaining certain distance from their repercussions.
 - » **Other factors** include lack of national capabilities, lack of resources, lack of national talent pool etc.
- **Examples of non-state actors:**

Types	Motives	Examples	Examples of their activities
State-Sponsored hacking groups	Advancing the states' geopolitical agenda as well as accruing financial gain	Lazarus Group: a North Korean-sponsored hacking organisation	A phishing attack on the Bangladesh Bank network to steal US\$81 million in 2016.
Cyber mercenaries	Financial gain	Team Jorge: A group of Israeli contractors engaged in malicious cyber activities	Kenya's presidential election campaign rigged by hacking personal chats and contacts of personal aides of President William Ruto to send messages to military commanders and ministers.
Hacktivists	Desire to contribute to social or political change	Ukraine IT Army: Cybersecurity experts who volunteered to attack Russia	Distributed denial of service attacks on 5,500 Russian websites in the initial months of the war.

- **Implications:**
 - » Cyberspace and non-state actors have reduced the gap between the well-off country and a developing country in terms of their capability to attack.
 - » Non state actors are also used for strengthening cyber defences.
- **Threats:**
 - » The non-state actors also have a very disruptive and destabilizing nature.

- » They can also be a threat to democratic process - for instance cyber mercenary group Team Jorge claims to have interfered in 27 Presidential-level campaigns worldwide through hacking, disinformation, planting fake intelligence etc.

- **What should be done?**

- » **International Treaty** - Against using non-state actors for cyber-attacks (use can be allowed for strengthening defences)
- » **Legislation** is necessary to curtail activities on non-state actors.

- **Conclusion:**

- » As geopolitical rivalries deepen with time, the use of non-state actors is only set to grow. It is important that the nation states come to a international agreement against the use of non-state actors.

12. BASICS: MAIN TYPES OF CYBER THREATS (TECHNICAL CLASSIFICATION)

1. **Malware** (Malicious Software) is any program of file that is harmful to a computer user. It includes computer viruses, worms, Trojan horses and spyware.
 - These malicious programs can perform a variety of functions, including **stealing, encrypting or deleting sensitive data**, altering or hijacking **core computing functions** and **monitoring users' computer activities**.
 - **Virus**
 - A virus is a malware that can execute itself and spreads by infecting other programs of file. Viruses are typically attached to an executable file or a word document. They often spread via P2P file sharing, infected websites, and email attachment downloads.
 - Once a virus finds its way onto your system, it will remain dormant until the infected host file or program is activated, which in turn makes the virus active enabling it to run and replicate on your system.
 - **Worm** is a type of malware that can **self-replicate without a host program**. Worms typically spread without any human interaction or directives from malware author.
 - **Note:** An important distinction between viruses and worms is that viruses require an active host program or an already infected and active operating system in order for viruses to run, cause damage and infect other executable files and documents. While, worms are stand-alone malicious program that can self-replicate and propagate via computer networks, without human help.
 - **Trojan Horse** is a malicious program that is designed to appear as a legitimate program. Once activated following installation, trojans can execute their malicious functions.
 - **Spyware** is a kind of malware that is designed to collect information and data (including intellectual property data) on users and observe their activity without users' knowledge.
 - **Ransomware** is a sophisticated malware that bypasses the traditional layers of security and makes the user's computer files inaccessible by either locking them up or encrypting them. The user is then asked to pay a "ransom" to the cybercriminals to regain access to the data.
 - E.g. **WannaCry** (May 2017)
 - Ransomware attacks on AIIMS in Dec 2022.
 - **Rootkit** is a type of malware designed to obtain administrator level access to the victim's system. Once installed, the program gives threat actor the root or privileged access to the system.

2. **Advanced Persistent Threat (APT)**

- An APT is a prolonged and targeted cyberattack in which an intruder gains access to a network and remains undetected for an extended period of time.
- The intention of an APT attack is usually to monitor network activity and steal data rather than to cause damage to the network or organization.
- They typically target organizations in sectors such as national defense, manufacturing and financial industry as these companies deal with high-value information, including IPR, military plans etc.

3. **Cryptojacking:**

- It is a cyber attack wherein a computing device is hijacked and controlled by the attacker and its resources are used to illicitly mine cryptocurrency.

4. **Denial of Service (DoS) - blocking access to websites**

- It is an attack in which a **malicious bot sends more traffic to a targeted IP address** that the programmers who planned its data buffers anticipated someone might send. The target becomes unable to resolve legitimate requests.

5. **Malware can spread through a number of methods**

- Physically** through a USB device
- Downloaded** from internet without users approval from malicious websites
- Phishing attacks/ Spear Phishing** are another common type of malware delivery mechanisms
 - **Phishing** is when malicious party send a fraudulent email disguised as a legitimate email, often purporting to be from a trusted source.
 - The message is meant to trick the recipient into sharing personal or financial information or clicking on a link that installs malware.
 - **Spear Phishing** is a phishing mechanism tailored for a specific individual or organization.
- Social Engineering** - is an attack vector that relies heavily on human interaction and often involves manipulating people into breaking normal security procedures and best practices in order to gain access to systems, networks or physical locations, or financial gains.
 - E.g. Honeytrap
- Exploits of zero day vulnerability**

Note: APT groups often use advanced attack methods such as spear phishing, social engineering and exploiting zero day vulnerability.

13. POST QUANTUM CRYPTOGRAPHY: SECURING DATA IN THE AGE OF CRYPTOGRAPHY

- **Example Question:**

- "As current security techniques like RSA and elliptic curves rely on mathematic problems vulnerable to quantum algorithms, the urgency to explore post-quantum cryptography alternatives becomes evident" Elaborate [10 marks, 150 words]

- **Introduction:**

- Current cryptography techniques such as RSA depends on Public Key Cryptography. They are securing most of the data today whether on physical hardware or the ones floating online. It is secure because the contemporary computers can't solve some problems like finding prime factors of large numbers. But, as Quantum computing with its almost infinite computing powers

develop, all these problems will be solved by Brute force and these Public key algorithm would become breakable **making the current security systems useless.**

- **Therefore**, governments across the world are rushing to develop quantum computing platforms and advanced security algorithms to defend against such machines.
 - For e.g., the US National Institute of Standards and Technology is working on **Post Quantum Cryptography Standardization Project.**
- **Post-Quantum Cryptography** involves exploring alternative techniques to counter vulnerabilities against quantum attacks.
- **What is India doing?**
 - India has recently announced National Quantum Mission.
- **Way Forward:**
 - More investment on R&D
 - More research labs and centre of excellence
 - Human Resource
 - Academia-Academia collaboration
 - Academia-Industry collaboration
 - Encouraging Private sector in the field
- **Conclusion:**
 - We are probably decades away from a quantum computer powerful enough to do anything meaningful or dangerous. However, experts believe that this is inevitable, and we will eventually develop such machines. Given the advantage it will give to first mover, it is important that we quickly and carefully transition to technologies secure against quantum attacks.